BEYOND EVALUATION USE

An Empirical Study of Involvement and Use/Influence in Multi-Site Evaluations

Report 10:
Survey Data Report for the
Math Science Partnership
Program Evaluation

Patricia A. Ross

University of Minnesota
College of Education and Human Development
Minneapolis, MN 55455

November 18, 2009

Project supported by the National Science Foundation Grant #REC0438545

Contributors

Principal Authors .....Patricia A. Ross
Principal Investigators ..... Jean King
.....Frances Lawrenz
Research Assistants.....Gina Johnson
..... Kelli Johnson
.....Denise Roseland

Table of Contents

INTRODUCTION **4**

METHODS **5**

MSP PROJECT LEADER SURVEY 5
ARCHIVAL REVIEW 7

MSP PROGRAM AND EVALUATION BACKGROUND AND CONTEXT **7**

MSP PROGRAM EVALUATION BACKGROUND 8
MSP PROGRAM EVALUATION QUESTIONS 8
MSP EVALUATION FACTS 8
MSP PROGRAM EVALUATION DESIGN AND COMPONENTS 9

PROJECT LEADER INVOLVEMENT AND USE **9**

LEVELS OF INVOLVEMENT 11
Table 3 Involvement – Planning *11*
PERCEIVED USE BY PROJECT LEADERS 13
Table 7 Influence on Knowledge and Understanding *13*
Table 8 Influence on Skills *14*
Table 9 Influence on Beliefs *14*
Table 10 Use of Knowledge, Products, Processes in Another Evaluation Context *15*
Table 11 Use of Findings *16*
Table 12 Influence of program evaluation on project leaders *16*
USE BY PROJECTS 17
INFLUENCE ON THE FIELD 18
Table 14 Influence on the Evaluation Field *18*
FACTORS AFFECTING USE 18
ENABLING FACTORS 18
LIMITING FACTORS 18

SUMMARY **19**

Introduction

In November 2004, the National Science Foundation (NSF) awarded Dr. Frances Lawrenz and Dr. Jean King of the University of Minnesota a three-year grant to study the effect that participation in program evaluations has on the subsequent use of evaluation results. This grant, entitled "Beyond Evaluation Use", examined multiple projects within each of four main NSF-funded programs. These programs included: Local Systemic Change (LSC), Collaboratives for Excellence in Teacher Preparation (CETP), Advanced Technological Education (ATE), and Mathematics and Science Partnerships (MSP).

The grant activities were aimed at measuring and describing how these four programs carried out their evaluation efforts. In particular, the research question was how these evaluation efforts involved the various projects in this process. A further question focused on whether involvement affected the ways in which projects, individuals involved with the projects, and the STEM education and evaluation communities used the results.

As part of the research project, an on-line survey addressing the program-level evaluations was administered with evaluators and project leaders. The purpose of that *first* survey was to investigate the interaction of use and involvement and the data from that survey has been reported in "Report 1: NSF Follow-up Survey Detailed Data Report." Detailed case studies for each of the four programs (LSC, CEPT, ATE and MSP-RETA), which examine each program's history and evaluation context, and discuss survey findings within the context of each program can be found in Reports 2 – 5.[1] A detailed case study of the MSP – RETA Program can be found in "Report 4: A Case Study of the Impact of the Utah State Math Science Partnership (MSP) – Research, Evaluation, and Technical Assistance (RETA) Project".

After the administration of the *first* survey, the Beyond Evaluation Use team decided that further research into the relationship between evaluation use and influence should be done. The team re-administered the survey to gather more data for a confirmatory factor analysis. As with the *first* survey, the *second* survey was sent to PIs, but also included co-PIs. This report summarizes the results of a *second* administration of the original survey of ATE, LSC, CEPT and MSP project leaders. The purpose of this second administration was to gather data for a confirmatory analysis. Since this data was available, however, a presentation and analysis of the results is being given here. As was done with the first survey, an analysis of the results is given to help explain the extent to which MSP projects believed they were involved in the planning and implementation of the evaluation, and how that involvement affected the use of the program findings.

The primary research questions guiding this inquiry are as follows:

1. What is the relationship between involvement in the MSP program evaluation and the use and influence of the evaluation on project-level staff and evaluators?
2. What factors other than involvement appear to be related to the use and influence of the program evaluation on the projects?

[1] http://www.cehd.umn.edu/projects/beu/ . See "Project Documents", Reports 2-5.

Methods

Data were collected through an online survey distributed to MSP project leaders. Additional data were collected through review of archival documents, publications, and reports produced by the MSP program and program evaluation.

MSP Project Leader Survey

The Beyond Evaluation Use team acquired names and email addresses for the principal investigators (PIs) and co-principal investigators (co-PIs) from the NSF website. Although the lists contained the most recent email addresses on record, some names were missing email addresses. In such instances, a member of the EUG team searched the Internet for an email address.

The PIs and co-PIs received an initial invitation email with a description of the survey, an explanation of its use, the Human Subjects approval number, and telephone numbers to call with questions. The email also contained a link to the online survey with a login and password. Two and five weeks following the first email, the team sent email reminders to non-respondents. The reminder email also contained a link to the online survey with a login and password. The data were collected from April 8, 2009 through May 31, 2009.

The second survey data is available in "Report 9: NSF Second Follow-up Survey Detailed Data Report". Survey findings presented in this report are taken from 57 close-ended items and four open-ended questions related to the involvement in and use of the MSP program evaluation. There was also an opportunity for respondents to provide additional comments in a final open-ended item.

A pre-notice email was sent to 42 of the 92 potential respondents, while 20 of the 92 received no pre-notice email. Based upon response rates, the research team found that response rates were higher for those respondents who had received a pre-notice email (50%) than for those that did not (40%).

Due to the fact that many of the email addresses were obtained via Internet search by the EUG team, only approximately 10 percent of the emails were returned as undeliverable. After receiving the initial email invitation, a number of potential respondents replied with valid reasons for why they should not be included in the survey, such as those on projects that had not yet begun the evaluation process or those who retired prior to their project being evaluated. These persons were removed from the sample. In total, 17 of the original 108 individuals contacted were either removed from the survey or had emails returned as undeliverable. Of these, 41 individuals replied resulting in a 45% response rate.

Team members reviewed responses to open-ended items. When appropriate, responses may be used to support or exemplify findings from the close-ended survey items or the project interview data, which are described in the next section of this report. The following table shows the

number and percentage of respondents who provided comments to each open-ended question or statement.

Items 59, 60, 64	Comment	No Comment	Percent Commenting
What major factors, if any, enabled the evaluation experience to influence you?	26	15	63
What major factors, if any, limited the influence the evaluation experience had on you?	15	26	37
Is there anything else you would like to share with us about your experiences with the program evaluation?	14	27	34

Nonrespondent study. To check for nonresponse bias, the team conducted a nonrespondent follow-up survey subsequent to the web survey of the PIs and co-PIs. Nonresponse error results when the people who responded to the survey are different from the individuals in the sample who did not respond, in a way that is relevant to the study. We developed a three-question nonrespondent survey to address the topics of involvement and use, as well as to gather some information about what kept individuals from responding to the initial survey and reminder emails. The nonrespondent survey items were:

1. How involved were you in the MSP Program Evaluation?
2. How much impact did the MSP Program Evaluation have on you?
3. What was the main thing that kept you from responding to our initial survey request?

A random sample of nonrespondents to the survey was drawn using Microsoft Excel. Each person was sent a personal, individually addressed email with a request for his or her input on the three items. Two reminder emails were sent to each of the respondents who did not respond to the initial request. Then, if email contact did not solicit a response, the individual was contacted by telephone. Forty-seven percent of the nonrespondents (14/30) completed the nonrespondent survey.

The results of the analysis comparing the MSP respondents and nonrespondents showed differences in the overall levels of involvement and use reported. The Involvement Mean of the survey respondents was higher (2.24) than the mean of the non-respondents (1.71). Similarly, the Use/Impact Mean was higher (2.54) for respondents than for non-respondents (1.71). The Involvement Mean for the respondents was calculated as the mean of thirteen questions on the original survey. The Use/Impact Mean was calculated in the same way, using the fourteen survey questions related to evaluation use and impact. The means for the non-respondents were simply the means of their answers to the two questions they responded to via email. Though these measurements are not identical, the two sets of means closely mirror one another in that the 4-point scale for each question or set of questions is the same.

The process of distilling the larger survey's multiple items related to use into a single question resulted in the choice of the word "impact" as a global concept to encompass questions about use of the process, instruments, and results of the evaluation. Therefore, the difference between respondents and nonrespondents may indicate the possibility of upward bias in the estimates of

overall use for MSP survey respondents or it may also be an artifact of the difference between the terms use, influence and overall impact.

Archival Review

The collection of archival documents for MSP was primarily conducted using electronic databases and Internet search engines. The NSF website provided information on the MSP program. The NSF documents included reports, grant solicitations, project summaries, and a "dear colleague" letter.

The primary source of information was the NSF-supported MSPnet[2], which is an electronic learning community that offers a centralized location for information on MSP projects and links to several other program-related websites. We also searched using Google, several electronic indexes including Academic Search, WilsonWeb (the H.W. Wilson Information Retrieval System for the World Wide Web), ERIC (the Education Resources Information Center), and the NSF-sponsored OERL (the Online Evaluation Resource Library).

MSP Program and Evaluation Background and Context

MSP Program

The NSF Math Science Partnership Program (MSP) began as a presidential initiative in 2001 and was authorized by Congress in 2002.[3] MSP is a major research and development effort that supports innovative partnerships to improve K-12 math and science instruction, enhance student achievement, and reduce the achievement gap in math and science performance among diverse student populations.[4]

MSP Program Facts

Period: 2002- present

Purpose: To integrate the work of higher education with K-12 to strengthen and reform mathematics and science education.

Primary Target: Universities, K-12 schools and corporations

MSP supports the development of partnerships to integrate the work of higher education and K- 12 STEM faculty. Typical project activities include new curriculum development and skill- building for teachers. Successful projects serve as models that can be widely replicated.[5] The partnerships must support the program's goals to provide challenging coursework; increase the quantity, quality, and diversity of math and science teachers; and establish results-oriented projects that implement evidence-based educational practices.[6] In 2006 MSP was bringing together 150 institutions of higher education, 550 school districts, and 3,300 schools, along with corporate partners including Pfizer, Ford, Texas Instruments, Inc., Xerox, IBM, Merck,

[2] http://hub.mspnet.org/index.cfm
[3] National Science Foundation Office of Inspector General. "Audit of NSF's Math and Science Partnership Program." OIG 04-2-003. May 14, 2004.
[4] NSF. 2006 MSP Program Solicitation. http://www.nsf.gov/funding/pgm_summ.jsp?pims_id=5756.
[5] NSF. 2006 MSP Program Solicitation. http://www.nsf.gov/funding/pgm_summ.jsp?pims_id=5756.
[6] National Science Foundation Office of Inspector General. "Audit of NSF's Math and Science Partnership Program."

and Intel.[7]

MSP Timeline and Budget

MSP awards competitive grants to teams of institutions: higher education, K-12, and other partners.[8] MSP projects emphasize innovation to improve K-12 math and science education. Since 2002, the MSP Program has made 80 awards totaling over $500 million across four program types[9]:

1. Comprehensive Partnerships implement change across the K-12 continuum in mathematics, science, or both.
2. Targeted Partnerships focus on improved student achievement in a narrower grade range or disciplinary focus in mathematics and/or science.
3. Institute Partnerships develop mathematics and science teachers as school- and district-based intellectual leaders and master teachers.
4. Research, Evaluation, and Technical Assistance (RETA) activities assist partnership awardees in the implementation and evaluation of their work.

MSP Evaluation Facts

NSF Grant Funding
•$500 million (2004-present)

NSF Goals for MSP Program Evaluation
The identification of educational strategies likely to have positive impacts on local school systems.

Evaluation Questions
1. How has MSP affected change in the teaching force, student achievement and other outcomes?
2. What factors have accelerated/constrained progress in MSP's achievements?
3. How have faculty from institutions of higher education participated in MSP?
4. What are MSP's features that are worth developing on a large scale?
5. What are the processes associated with these features?
6. What are the conditions associated with the demonstrated quality of MSP?

Personnel/Expertise
Robert K. Yin (PI), COSMOS Corporation in partnership with George Mason University and Brown University. Co-PIs: Jennifer Scherer (COSMOS), Patricia Moyer-Packenham (USU, formerly GMU) and Kenneth Wong (Brown).

MSP Program Evaluation Background

The Math and Science Partnership Program Evaluation (MSP-PE) began in 2004. After a two-year period in which the evaluation was designed and planned, and necessary clearances were obtained, the evaluation activities began in 2006. Through a study of the program's main themes and early activities, COSMOS Corporation developed an evaluation framework. The details of this study can be found in Yin (2008). [10]

MSP Program Evaluation Questions

The evaluation questions for the MSP program evaluation grow out of three questions and three topics that were specified in the original NSF solicitation for a program evaluation. They include:

1. How has the MSP Program affected, influenced, or been associated with changes in the K-12

[7] NSF. MSP At a Glance. http://nsf.gov/pubs/2005/nsf05069/nsf05069.pdf., p.12.
[8] NSF. MSP At a Glance. http://nsf.gov/pubs/2005/nsf05069/nsf05069.pdf., p.12.
[9] http://www.nsf.gov/ehr/MSP/nsf05069_3.jsp
[10] Yin, R. K. (2008). The Math and Science Partnership Program Evaluation: Overview of the first two years. *Peabody Journal of Education*, *83*, 486 – 508.

mathematics and science teaching force, K-12 student achievement in math and science, and other outcomes associated with the program?
2. What factors or attributes appear to have accelerated or constrained progress in the MSP Program's achievements?
3. How have disciplinary faculty (math, science, and engineering) from institutions of higher education participated in the MSP Program, and what has been their role in the program's achievements?
4. The MSP's Program features, including MSP-related discoveries and innovations in math and science education worth developing on a large scale.
5. The processes influencing, interfering, or associated with the outcomes of the features.
6. The conditions associated with the demonstrated quality and innovativeness of the MSP Program.[11]

MSP Program Evaluation Design and Components

Through an assessment of the MSP Program, the need for a multi-institutional framework for the evaluation was established. Details of the program assessment and development of the evaluation design are provided in "The Math and Science Partnership Program Evaluation: Overview of the First Two Years" (Yin, 2008). Within this multi-institutional framework, core participants include K-12 school systems and institutions of higher education (K-20 education). However, other key institutional partners of relevance include state education agencies, informal science education organizations, business and industry, community organizations and others with a stake in educational excellence. The actions of these institutions are seen as related through a series of pathways. For example, students enter through K-12 educational institutions. After graduation, they move into the work force (business and industry) (one pathway) or undergraduate programs at the institutions of higher education (another pathway). Some may become science and math teachers at K-12 institutions, or enter graduate programs at institutions of higher education, then go on to become scientists, mathematicians or engineers. The institutions can be seen as being inter-related by multiple pathways between them. This inter-relatedness of institutions, and the people moving among them, suggested a systems framework for the evaluation. The emphasis with such a framework is upon the interactions of the institutions, and the multi-institutional transitions that occur.

Further, such a framework suggests a sub-study strategy. Sub-studies allow for a series of smaller studies, each with its own design and purpose, focused upon a different part of the system. To synthesize evidence concerning the MSP Program overall, meta-analytic strategies (both qualitative and quantitative) are used. Thus, for example, one sub-study is K-12 student achievement trends. Another sub-study involved an examination of awardees' annual reports. Case studies of selected MSP projects are also being conducted. Later phases of the evaluation will incorporate all of this original field data.

Project Leader Involvement and Use

[11] Yin, R. K. (2008). The Math and Science Partnership Program Evaluation: Overview of the first two years. *Peabody Journal of Education, 83*, 486 – 508.

This section presents data from the survey to describe how project leaders (i.e., PIs and co-PIs) reported being involved in and influenced by the MSP program evaluation. This presentation of findings is organized around six topics that guided the data collection and analysis for the study. For each topic, quantitative results are discussed. The six topics are:

- Motivation for Involvement
- Levels of Involvement
- Perceived Effect of Involvement
- Use by Project Leaders and Evaluators
- Use by Projects
- Factors Affecting Use

Motivation for Involvement

Item 4 of the survey addressed the motivation that PIs and co-PIs had, at different stages (planning, implementation and communication) of the evaluation, for being involved. The planning stage includes activities such as discussing the focus of the evaluation, identifying evaluation planning team members, developing the evaluation plan, while the implementation stage includes developing data collection instruments and processes, collecting, analyzing, reviewing, and interpreting data. The final stage of the program evaluation – the communication stage – is comprised of activities such as developing future plans for your project, writing and reviewing evaluation reports, presenting evaluation findings. Among the three stages of the evaluation, findings suggest that notable difference can be seen in the motivation for involvement. In particular, during the planning stage of the program evaluation respondents claimed to be less likely to be voluntarily involved (39.3%) than involved because of either a requirement or being asked to participate (60.7%). During the implementation stage, we saw an evening out of this difference, with respondents claiming to be about equally likely to be either volunteers (48.4%) or required/asked (51.7%). Finally, during the communication stage, motivation appears to be more likely voluntary (57.1%), than required or asked (42.9%).

If we are to consider the differences in external motivation (required vs. asked), we can see that approximately a third of respondents participate at all stages because of a perceived requirement. What is most notable in terms of external motivation for involvement is the precipitous decline, over the course of the program evaluation, in participation due to being asked. While nearly a third of respondents were involved because of being asked during the planning stage, only 1 of the 28 respondents claimed to be involved because of being asked at the communication stage.

Table 2 *Motivation for Involvement*

Item 4	I was required	I was asked	I volunteered	N
Planning stage	32.1%	28.6%	39.3%	28
Implementation stage	32.3%	19.4%	48.4%	31
Communication of evaluation findings stage	39.3%	3.6%	57.1%	28

Levels of Involvement

The survey asked respondents about their perceptions of involvement during the planning, implementation and communication phases of the program evaluation. Responses to the involvement questions are summarized in Tables 3, 4 and 5. Each involvement question began with "I was involved in…."

Table 3 *Involvement – Planning*

Items 1-3	No	Yes, A little	Yes, Some	Yes, Extensively	N	*M*	*SD*
Discussions that focused the evaluation	22.5%	10.0%	17.5%	50.0%	40	3.50	1.24
Identifying evaluation planning team members	37.8%	10.8%	21.6%	29.7%	37	2.43	1.28
Developing the evaluation plan	30.8%	12.8%	25.6%	30.8%	39	2.56	1.23

Table 4 *Involvement – Implementation*

Items 5-10	No	Yes, A little	Yes, Some	Yes, Extensively	N	*M*	*SD*
Developing data collection instruments	52.5%	17.5%	17.5%	12.5%	40	1.90	1.10
Developing data collection processes	45.0%	30.0%	15.0%	10.0%	40	1.90	1.01
Collecting data	45.0%	17.5%	25.0%	12.5%	40	2.05	1.11
Reviewing collected data for accuracy and/or completeness	55.0%	20.0%	12.5%	12.5%	40	1.83	1.08
Analyzing data	43.6%	25.6%	15.4%	15.4%	39	2.03	1.11
Interpreting collected data	30.0%	22.5%	22.5%	25.0%	40	2.43	1.17

Table 5 *Involvement – Communication*

Items 12-15	No	Yes, a little	Yes, some	Yes, extensively	N	M	SD
Writing evaluation reports	59.0%	17.9%	15.4%	7.7%	39	1.72	.999
Reviewing evaluation reports for accuracy and/or completeness	28.9%	23.7%	18.4%	28.9%	38	2.47	1.20
Presenting evaluation findings (e.g., to staff, to stakeholders, to an external audience)	38.5%	25.6%	20.5%	15.4%	39	2.13	1.10
Developing future project plans based on evaluation results	23.1%	20.5%	12.8%	43.6%	39	2.77	1.24

What can be seen from these responses is a notable lack of involvement during the implementation phase. Except for being involved with interpreting collected data, nearly half of all respondents claimed to have no involvement in any of the other implementation activities. On the other hand, over half of all respondents maintained that they were either extensively or somewhat involved with planning activities. Half of all respondents claimed to be involved extensively in discussion that focused the evaluation.

The communication stage of the evaluation saw the most variable responses in terms of how respondents recalled their involvement in the activities of this stage. In particular, respondents

were overwhelmingly not involved (59%) or only a little involved (17.9%) in writing evaluation reports. However, the majority of respondents (47.3%) were either extensively or somewhat involved in reviewing these reports. And while involvement in presenting evaluation finding was distributed, once again toward no involvement or little involvement, it was somewhat more evenly distributed among respondents than involvement in writing reports. Activities involving the development of future projects plans, much like reviewing evaluation reports, saw more involvement with over half (56.4%) of respondents claiming to have been either extensively or somewhat involved in this activity. This division that we see in involvement in communication activities (writing reports and presenting findings vs. reviewing reports and developing future plans) can most easily be explained by the fact that PIs and co-PIs would not be directly writing the evaluation reports. The writing of reports was the responsibility of the evaluator. However, PIs and co-PIs would be involved in reviewing these reports and, perhaps using the results presented therein.

Perceived Effect of Involvement

Questions 17 through 22 concerned respondents' perceptions concerning the influence their input had on each stage of the program evaluation. In providing this feedback, respondents were asked to differentiate between having their input taken into consideration and having their input actually used. Survey results suggest that, of the PIs and co-PIs who were involved in the various stages of the evaluation and had some sense of how their input was perceived, these respondents overwhelmingly believed that their input was both considered and used. In all except one case ("Input during implementation was used") no respondent believed that their input went unconsidered or unused.

What is noteworthy in this data is the consistency in response rates between the three questions concerning "consideration" and those concerning "use". For each stage of the evaluation we find that a similar number of respondents perceived their input as being both considered and used. This effect may be due to the responsiveness of the evaluators not only in thinking about the input provided but also in finding ways to employ that input.

Table 6 *Influence on the Evaluation*

Items 17-22	Was not involved	No	Yes, a little	Yes, some	Yes, extensively	Do not know	N
Input during *planning* was taken into consideration	25.0%	.0%	12.5%	27.5%	32.5%	2.5%	40
Input during *implementation* was taken into consideration	26.8%	.0%	14.6%	24.4%	29.3%	4.9%	41
Input during *communication* was taken into consideration	24.4%	.0%	19.5%	24.4%	26.8%	4.9%	41
Input during *planning* was used.	24.4%	.0%	12.2%	24.4%	34.1%	4.9%	41
Input during *implementation* was used.	24.4%	2.4%	14.6%	26.8%	26.8%	4.9%	41
Input during *communication* was used.	24.4%	.0%	22.0%	22.0%	24.4%	7.3%	41

Perceived Use by Project Leaders

The survey asked project leaders to reflect on the extent to which the program evaluation influenced their knowledge/understanding, skills and beliefs. Additionally, project PIs were asked about their use of the program evaluation process and findings in another evaluation context. Each of these is examined in turn.

.

Influence on Knowledge and Understanding

Respondents most commonly believed that their knowledge and understanding of evaluation (all stages), STEM education and evaluation, and their own project increased "some" as a result of the program evaluation. While the program evaluation appears to have most strongly influenced respondents' knowledge and understanding of the respondent's own project, how to plan an evaluation, how to implement an evaluation and understanding STEM education evaluation were all similarly influenced for the majority of project leaders. With respect to STEM education, respondents reported that the program evaluation was less likely to increase their knowledge and understanding (M = 2.30, SD = 1.04) than it was to influence other areas. Nearly a third (30%) of respondents said that the program evaluation had no influence on their knowledge and understanding of STEM education.

Interestingly, however, one respondent did take the time to comment about the positive influence of the evaluation and, in particular, on her own teaching. "It was the first time I had learned about evaluation and I found it interesting. Perhaps most importantly, I was in seminars about Formative Assessment that I then put to use in my own teaching."

Table 7 *Influence on Knowledge and Understanding*

Items 25-30	No	Yes, a little	Yes, some	Yes, extensively	N	M	SD
The evaluation increased my knowledge/ understanding of how to plan an evaluation.	22.0%	12.2%	39.0%	26.8%	41	2.71	1.10
The evaluation increased my knowledge/ understanding of how to implement an evaluation.	19.5%	12.2%	46.3%	22.0%	41	2.71	1.03
The evaluation increased my knowledge/ understanding of how to communicate evaluation findings.	22.0%	19.5%	39.0%	19.5%	41	2.56	1.05
The evaluation increased my knowledge/ understanding of STEM education.	30.0%	22.5%	35.0%	12.5%	40	2.30	1.04
The evaluation increased my knowledge/ understanding of STEM education evaluation.	19.5%	14.6%	43.9%	22.0%	41	2.68	1.04
The evaluation increased my knowledge/ understanding of my project.	19.5%	9.8%	34.1%	36.6%	41	2.88	1.12

Influence on Skills

Table 8 presents respondents' thought on how the program evaluation influenced their skills. Although the program evaluation's influence on skills reflects a similar pattern to its influence on

knowledge and understanding, some notable differences exist. Fewer respondents, in all categories of influence, felt that the evaluation had influenced them extensively. What is interesting is that in some categories (skills in communicating finding, skills as a STEM education evaluator) even though they believed their knowledge and understanding was improved, respondents maintained that their skills were only influenced "a little".

Table 8 *Influence on Skills*

Items 31-36	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*
The evaluation improved my skills in planning an evaluation.	17.5%	17.5%	47.5%	17.5%	40	2.65	.975
The evaluation improved my skills in implementing an evaluation	19.5%	29.3%	39.0%	12.2%	41	2.44	.950
The evaluation improved my skills in communicating evaluation findings.	22.0%	31.7%	39.0%	7.3%	41	2.32	.907
The evaluation improved my skills as a STEM educator.	36.6%	24.4%	26.8%	12.2%	41	2.15	1.06
The evaluation improved my skills as a STEM education evaluator.	17.1%	29.3%	43.9%	9.8%	41	2.46	.897
The evaluation improved my skills for working on my project.	22.0%	14.6%	43.9%	19.5%	41	2.61	1.04 6

Influence on Beliefs

In much the way that respondents were asked to reflect upon the program evaluation's influence on their knowledge, understanding and skills, respondents were asked to also reflect upon the influence that the evaluation had on their *beliefs* about the importance of the evaluation. Table 9 presents these results.

Table 9 *Influence on Beliefs*

Items 38-43	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*
The evaluation increased my belief in the importance of planning an evaluation.	22.0%	12.2%	29.3%	36.6%	41	2.80	1.17
The evaluation increased my belief in the importance of implementing an evaluation.	22.0%	12.2%	29.3%	36.6%	41	2.80	1.17
The evaluation increased my belief in the importance of communicating evaluation findings.	31.7%	9.8%	22.0%	36.6%	41	2.63	1.28
The evaluation increased my belief in the importance of STEM education.	48.8%	17.1%	22.0%	12.2%	41	1.98	1.11

Items 38-43	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*
The evaluation increased my belief in the importance of STEM education evaluation.	29.3%	9.8%	43.9%	17.1%	41	2.49	1.10
The evaluation increased my belief in the importance of my project.	24.4%	17.1%	34.1%	24.4%	41	2.59	1.12

Use in another Context

Questions 45 – 49 sought information from respondents concerning the use of the evaluation knowledge, products, or processes in another (subsequent) evaluation context. The survey directed only respondents who had participated in a subsequent evaluation to respond to these questions. If a respondent had not participated in another evaluation, they were instructed to continue with question 50.

For those respondents who participated in another evaluation after the MSP program evaluation, we see that what was learned from planning (M = 2.9, SD = 1.15) and what was learned from implementation (M = 2.82, SD = 2.82) were the two areas most strongly affected by the evaluation. However, over half of all respondents claimed at least some use of past knowledge, products or processes in another evaluation context. Data collection instruments stood out as one category where approximately a third of the respondents felt they did not use the evaluation product in another context.

Table 10 *Use of Knowledge, Products, Processes in Another Evaluation Context*

Items 45-49	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*
I used what I learned from planning the program evaluation in another evaluation.	18.2%	13.6%	27.3%	40.9%	22	2.91	1.15
I used the evaluation plan from the program evaluation as a model in another evaluation.	22.7%	18.2%	27.3%	31.8%	22	2.68	1.17
I used what I learned from implementing the program evaluation in another evaluation.	18.2%	22.7%	18.2%	40.9%	22	2.82	1.18
I used data collection instruments from the program evaluation in another evaluation.	31.8%	13.6%	36.4%	18.2%	22	2.41	1.14
I used what I learned from communicating evaluation findings from the program evaluation in another evaluation.	27.3%	18.2%	31.8%	22.7%	22	2.50	1.14

Use of Findings

Questions 50 – 58 asked PIs and co-PIs to report the ways in which they used the program evaluation findings. Table 11 reports these results. One notable outcome of this series of

questions is that respondents claimed to use the program evaluation findings to develop future plans for their projects at a slightly higher rate (M = 2.71, SD = 1.15) than for any other purpose. But once again, this was a small increase in average response and, in general respondents appear to be fairly equally distributed in their use of evaluation findings. Across all categories of use, approximately a third of respondents did not use the program evaluation finding, a little more than a third used them "a little" or "some", while just under a third used them "extensively".

Table 11 *Use of Findings*

Items 50-58	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*
I used the program evaluation findings to make decisions about the future existence of my project (e.g., cancel or continue).	29.3%	9.8%	36.6%	24.4%	41	2.56	1.16
I used the program evaluation findings to develop future plans for my project.	24.4%	9.8%	36.6%	29.3%	41	2.71	1.15
I used the program evaluation findings to increase the attention given to my project.	34.1%	12.2%	31.7%	22.0%	41	2.41	1.18
I used the program evaluation findings to meet contractual and/or legal requirements of my project.	34.1%	7.3%	24.4%	34.1%	41	2.59	1.28
I used the program evaluation findings to enhance my organization's commitment to my project.	34.1%	14.6%	34.1%	17.1%	41	2.34	1.13
I used the program evaluation findings to seek funding for my project(s).	36.6%	4.9%	36.6%	22.0%	41	2.44	1.21
I used the program evaluation findings to promote additional evaluation of my project(s).	39.0%	9.8%	29.3%	22.0%	41	2.66	1.20
I used the program evaluation findings to make changes to my project(s).	29.3%	4.9%	36.6%	29.3%	41	2.49	1.16
I used the program evaluation findings to help plan a new project(s).	31.7%	9.8%	36.6%	22.0%	41	2.50	1.03

Comparisons Among the Types of Influence

As shown in Table 12, data from the survey suggest that the MSP program evaluation had the greatest influence on project leaders' use of knowledge, products, and/or processes in other contexts (M = 2.66, SD = 1.07). However, since this number is based upon only those reporting having conducted a subsequent evaluation (N = 22), we need to be careful interpreting this result in comparison with the others. The influence that the program evaluation had on all of the other areas was notable. The area where influence was least was is the area of skills. However, respondents still, on average, reported somewhat more than a little influence even in this area.

Table 12 *Influence of program evaluation on project leaders*

Type of Influence	N	*M*	*SD*
Influence on knowledge/understanding (q25-30)	40	2.61	.933
Influence on skills (q31-36)	40	2.47	.834
Influence on beliefs (q38-43)	41	2.55	1.01
Use of knowledge, products, processes in other contexts (q45-49)*	22	2.66	1.07
Ways of using findings (q50-58)**	41	2.50	1.03

*items asked only of those reporting having conducted a subsequent evaluation.
**items asked only of project PIs.

One interesting feature of this data is the difference that we find between perceived "influence on knowledge/understanding", on the one hand, and "influence on skills" on the other. One possible explanation for this may simply lie in the fact that only approximately half of the respondents went on to perform another evaluation. While respondents felt their knowledge and understanding was influenced, without an opportunity to put this knowledge into practice, half of respondents were unable to accurately assess the impact on their skills.

Use by Projects

As we saw above, survey items focused on the ways in which project leaders were influenced by the program evaluation, including the extent to which they used the process and findings to make decisions about projects. Some of the open-ended responses allowed for deeper understanding about the ways in which the program evaluation influenced the projects. One important caveat concerning the open-ended items concerns the fact that some of the respondents answered these questions in reference to their own project's evaluation, as opposed to the larger program evaluation.

In spite of this limitation, the evaluation team examined open-ended items carefully with this possible misunderstanding in mind, and found that several themes emerged relating to the influence of the program-level evaluation on respondents. One theme concerned failures of communication. Respondents noted that a lack of communication of both findings of the MSP program evaluation and opportunities to participate in the evaluation were what impeded use of findings for their own projects. As one respondent noted:

> I collected and supplied data for my project as requested for the overall program evaluation. We also collected, analyzed, and reported data specific to our individual project. The latter, local evaluation has been very important to the project. The MSP Program-wide evaluation hasn't had much, if any influence on our work. I'm not sure I'm aware of the findings. If I were, they might be influential.

Another respondent echoed this concern. "…some great opportunities were envisioned for …[PIs] to be engaged in the [sic] MSP-wide evaluation. Yet somehow as an active co-PI, I was

not aware of these opportunities. So I would recommend a different communication strategy next time."

Influence on the Field

Questions 61 and 62 looked at respondent's assessment of how influential the MSP program evaluation was on the STEM education community and the evaluation community. Tables 13 and 14 summarize these survey responses.

Table 13 *Influence on STEM Education Community*

Not influential at all	Marginally influential	Somewhat influential	Influential	Very influential	Total
12.5%	22.5%	37.5%	22.5%	5.0%	40

Table 14 *Influence on the Evaluation Field*

Not influential at all	Marginally influential	Somewhat influential	Influential	Very influential	Total
17.5%	30.0%	30.0%	17.5%	5.0%	40

While similarities exist in the distributions of responses for these two questions, overall the influence on the field of evaluation was seen as being somewhat lower than the influence on the STEM community. Nevertheless, the trend among respondents was to see the influence of the evaluation on these two areas as "somewhat" influential.

Factors Affecting Use

Survey respondents were asked to identify any major factors that either enabled or limited the influence of the evaluation. We have already discussed the effect that lack of communication was thought to have on the projects. However, having experienced evaluators was identified as enabling the evaluation. Respondents noted, however, concerns regarding the lack of connection with local-level work, and the seeming "bookkeeping" nature of the evaluation.

Enabling Factors

While lack of communication may have hindered evaluation use, the overall role of evaluators', including their communication and guidance through the evaluation process, was highlighted as a factor that facilitated understanding, and in turn encouraged use. One respondent noted "expert evaluators led me through the process from logic models through data collection, to analysis and crafting reports and communications." Another noted "discussions and back and forth communications with the evaluators were quite helpful."

Limiting Factors

The evaluation work done for the program-level evaluation was occasionally seen as disconnected to the work being done at the local level. One respondent wrote "As the PI of a RETA grant, the MSP evaluation was not very connected to my work." Another claimed, "I don't think we've been very heavily influenced except to get ready for a site visit and to assemble and organize the data that's been requested of us."

Summary

This report describes the ways in which MSP project leadership and evaluators felt involved in the MSP program evaluation and subsequently used the results.

Among the three stages of the evaluation, findings suggest that a notable difference can be seen in the motivation for involvement. In particular, during the planning stage voluntary involvement was lowest. However, during the implementation stage it increased to the same level as being required or asked, and became the most common motivation during the communication stage. There was also significant difference in the involvement at different stages. There was a notable lack of involvement during the implementation stage, but over half of all respondents maintained that they were either extensively or somewhat involved with planning activities. Respondents' recollection of their involvement in the activities of the communication stage varied the most.

Survey results suggest that PIs and co-PIs involved in the various stages of the evaluation and with some sense of how their input was perceived overwhelmingly believed that their input was both considered and used. In addition, project leaders most commonly believed that their knowledge and understanding of evaluation (all stages), STEM education and evaluation, and their own project increased "some" as a result of the program evaluation. This also held for the influence on their skills, though fewer respondents, in all categories of influence, felt that the evaluation had influenced their skills extensively.

Concerning the extent to which project leaders used the process and findings to make decisions about projects, one theme that emerged was the overwhelmingly positive role of the collaborative effort that the evaluation involved. The back and forth discussion with evaluators, open and excellent communication and leadership were features of this collaboration that were repeatedly noted. In addition to collaborative efforts, the positive effect of experienced evaluators was also a common positive factor identified as enabling the evaluation. In particular, evaluators' expertise, critical-mindedness, ability to lead PIs and co-PIs through the evaluation process and establishment of a professional culture were also identified as enabling the evaluation. Respondents noted, as well, frustrations and issues that arose with tools and instruments used in this evaluation. This concern may be the primary limiting factor of the effect of the MSP evaluation.